

DOCUMENTATION FOR THE RATPOINTS PROGRAM

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1 Introduction

This paper describes the `ratpoints` program. This program tries to find all rational points within a given height bound on a hyperelliptic curve in the most efficient way possible.

2 History and Acknowledgments

This program goes back to an implementation of the ‘quadratic sieving’ idea by **Noam Elkies** that was around in the early 1990s. My own first contribution was to replace the char arrays that were used to store the sieving information by bit arrays, in 1995. **Colin Stahlke** then made use of the `gmp` library, so that points could be checked exactly, and implemented the selection of sieving primes according to their likely success rate, in 1998. After that, I successively put in numerous improvements (and some bug fixes). For details of how the program works, see Section 7 below.

Along with **Noam Elkies** and **Colin Stahlke**, I would like to thank **John Cremona** and **Sophie Labour** for bug reports and suggestions for improvements. Further thanks are due to **Bill Allombert** for a first implementation of the use of 256-bit (and possibly 512-bit) registers.

From September 2026, I have been using **Claude Code** (in various incarnations: Opus 5, Fable 5.1) to optimize the code and find and fix bugs. This has resulted in the current version 3.0.0, which includes a whole array of optimizations compared to the 2.2.x versions, making the final program run 2 to 10 times faster now (depending on the kind of curve and the height bound). Based on this aggregate very noticeable improvement and the change in the development workflow, I decided to increase the major version number from 2 to 3, even though many of the optimizations are incremental and there are few really new ideas involved. The next version, v3.1.0, will include multi-threading abilities. Stay tuned!

Note that version 2.2.4 is the last one that supports 32-bit architectures; versions 3.0.0 and up need a word size of 64 bit.

3 Availability

The `ratpoints-3.0.0` package can be downloaded from my homepage, see [rat], or obtained from github at <https://github.com/MichaelStollBayreuth/ratpoints>.

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4 Installation

This section describes the installation procedure under Linux.

4.1 Extract the archive

```
> tar xzf ratpoints-3.0.0.tar.gz
```

This sets up a directory `ratpoints-3.0.0` containing the various files that belong to the installation. Alternatively, clone the github repository mentioned above into a directory of your choice (and replace `ratpoints-3.0.0` by the name of that directory in the instructions below).

4.2 Build the program and library

Do

```
> cd ratpoints-3.0.0
> make all
```

This will build the library `libratpoints.a`, the executable `ratpoints`, and also run `pdflatex` to generate this documentation.

Five of the constants that govern the sieve depend on the machine rather than on the curve. Reasonable values are compiled in, but `make tune` will measure better ones for your machine if you want it to; see Section 8.4. It is a separate step because it takes several minutes.

By default, the program has the 53 odd primes below 256 available and considers the smallest 30 of them; for a curve that needs more, it will look further of its own accord, as described in Section 8.1. If you intend to do computations with curves that have many rational points, it may make sense to make more primes available. This can be achieved via

```
> make all PRIME_SIZE=9
```

The `PRIME_SIZE` argument can be given any value from 5 to 10; the default value is 8. The precise meaning is that the program will work with primes $< 2^s$, when `PRIME_SIZE = s`. The Makefile notices that the setting has changed and rebuilds what depends on it. Only the primes a run actually asks for get sieving tables, so a larger table costs nothing until it is used (Section 5.5).

The sieve works on several numerators at a time, in a register whose width is selected by the line `CCFLAGS1 = ${CCFLAGS256}`

in the Makefile. It can be set to `${CCFLAGS64}` (64-bit registers, i.e., plain unsigned longs, which works on any machine the library builds on at all), `${CCFLAGS128}`, `${CCFLAGS256}` (the default, using 256-bit AVX registers) or `${CCFLAGS512}`. The Makefile notices that the setting has changed and rebuilds everything that depends on it, so a `make distclean` in between is not necessary. Wider registers help the first phase of the sieve, where most of the time goes on a curve with few rational points, but they help the second phase much less and the sieve tables not at all, so the gain is nowhere near proportional to the width: on the author's machine, 64, 128 and 256 bits take 21.3, 15.2 and 12.2 units of time on such a curve.

You may have to change `-mavx2` in the line

```
CCFLAGS256 = -DUSE_AVX -mavx2
```

into `-mavx` or leave it out altogether, depending on the capabilities of your CPU. You will notice that AVX2 (or AVX) is not supported, when the executable throws an 'unknown instruction' error or similar.

There is also a variant using 512-bit registers, activated by

```
CCFLAGS1 = ${CCFLAGS512}
```

which requires a CPU with AVX512F capability. Note that this variant has so far only been tested indirectly (see the change log entry below), since I have no such CPU available; you should therefore run `make test` before relying on it. It is also worth timing the 256-bit and the 512-bit version against each other on your machine: the sieving loop is limited by how fast it loads bit arrays from the first-level cache and by how much of that cache the sieve tables occupy, rather than by arithmetic, so wider registers do not automatically pay off, and some Intel server processors reduce their clock frequency when 512-bit instructions are used. The one data point available so far is encouraging, though: on an AMD Ryzen 9 9950X3D (Zen 5 architecture), **Drew Sutherland** measured a speed-up of 13 to 14% over the 256-bit version, for a curve with many rational points on version 2.2.3.

If you leave out `-mavx512f` and compile with just `-DUSE_AVX512`, then `gcc` will express the 64-byte vectors through narrower operations, so that the code runs on any machine. This is of no use in production (it is slower than the 256-bit version), but it does allow the 512-bit code path to be tested without the corresponding hardware. `gcc` will warn that passing a 512-bit vector argument without AVX512F enabled changes the ABI; this is harmless, as the whole library is built with the same flags.

4.3 Run various tests

Run

```
> make test1
```

in the working directory. This will build an executable `rptest` and then run (and time) it. Finally, the output (which was written to a file `rptest.out`) is compared against `testbase`, which contains the output of a sample run. The two should be identical; otherwise the message `Test failed!` will be displayed (on its own on a line) and the target fails, so that `make` exits with an error and names the test.

```
> make test1once
```

runs the same curves again, but with the input fields of the argument structure set once before the loop over the curves instead of once per curve, as a program using the library may do since version 3.0.0 (Section 6); `rptest` prints a line whenever the library has changed one of those fields, and the output is compared against `testbase` as before.

```
> make test1many
```

does what `make test1` does, but with a second collection of curves, in `testdata-many.h`, whose reference output is `testbase-many`. These curves have (comparatively) many rational points, which makes them sieve quite differently from the random curves in `testdata.h`: on a random curve about one numerator in 10^4 survives the first sieving stage, on these up to a hundred times more, so the best division of labour between the two stages is a different one. The two tests take about the same time and between them cover the two regimes that occur in practice; both should be used whenever the parameters are retuned, since a setting that suits one of them can be a poor choice for the other. That is what `make tune` does with them; see Section 8.

```
> make testhigh
```

```
> make testhighmany
```

run the same two regimes at a height bound of $2 \cdot 10^5$ rather than the 16383 of `test1` and `test1many`, and take about a minute each. The height bound decides a good deal of what a test measures. A sieve table is built the first time a given prime meets a given class of denominators and is reused afterwards, so the total cost of building the tables is bounded by the number of primes and does not grow with the height, whereas the sieving does: on the random curves the sieve tables are 4% of `make test1` but 0.3% of `make testhigh`, the whole of the set-up per denominator 8.5% against 1.3%, and the two sieving phases themselves go from 57% to 90% of the run. These are therefore the tests to judge a change to the sieving loops by. `testhigh` uses the same curves as `test1`; none of them has a rational point of height between 16383 and $2 \cdot 10^5$, so the expected output is `testbase` again. `testhighmany` uses the thirty curves of `testdata-many.h` that run out of sieving primes, collected in `testdata-high-many.h` with reference output `testbase-high-many`.

```
> make test2
```

calls `ratpoints` on a curve with lots of rational points and with a fairly large height bound, times it, and compares the output to what is expected.

```
> make test3
```

runs the invocations of `ratpoints` collected in `test3.sh` (which check that the bugs fixed in versions 2.2.4 and 3.0.0 stay fixed) and compares the output against `testbase3`.

```
> make test4
```

runs a few hundred invocations of `ratpoints` collected in `test4.sh`, chosen so that every branch of the code runs at least once: degrees 1 to 9 and up to the maximum of 100, odd and even degrees, square and non-square leading and constant coefficients, every reason to reverse the polynomial, curves with no points modulo some prime or no admissible numerator modulo 64, forbidden divisors of every kind, every option, restricted denominator ranges and search intervals, height bounds from 1 to $2^{63} - 1$ (those come with a denominator range of one and a narrow search interval, so that the run stays short), and every error message. The output is compared against `testbase4`; the whole test takes about a second. Before each invocation the script prints its arguments, so that a failing test can be found from the difference between `test4.out` and `testbase4`. The reference was checked independently of the sieve: `verify-test4.py` reads it and, for every invocation that prints points, searches every coprime pair (a, b) in the range by brute force, in arbitrary precision; it can be run again whenever the reference changes (it needs Python 3, and PARI/GP for the 19 searches, on two curves, that are too large for it).

```
> make testapi
```

runs `rpapi`, a program that tests what the library offers and the command-line program cannot reach: the checks that `find_points_work` makes on its arguments and the error codes it returns, a callback that stops the search or declines a point, the flags, and a sequence of searches on one initialized structure (Section 6); the output is compared against `testbase-api`.

```
> make test4configs
```

runs `test4` again on the library built with each of the other compile-time switches: the register widths 64, 128 (both variants) and 512 (emulated); `RATPOINTS_CHUNK=1`, one register in the first phase; `USE_LONG_IN_PHASE_2`, the second phase one word at a time; `PRIME_SIZE 7`, the smaller prime table; and composite moduli off and unbounded. Each is built in a directory of its own, so that the build in the working directory is left alone (`test4-configs.sh`); the points found must not depend on any of these, so every build is compared against the same `testbase4`. It takes about a minute. The script exits with status 1 when an output differs from the reference and with 2 when a build failed; so does `coverage.sh` below.

```
> make coverage
```

measures how much of the code the tests execute, with `gcov`: it builds the sources instrumented and unoptimised in `build-coverage/`, runs the tests through that build (all of `make test` but `test2` and `timing`, which add nothing but minutes) and prints, for each source file, the lines executed and the branches taken at least once (`coverage.sh`); the annotated sources are left in that directory. When this was written the tests executed 99.9% of the lines of `find_points.c`, 96% of `sift.c`, all of `init.c` and `sturm.c` and 99% of `main.c`; what is left is guards against states no caller can produce, arms that other compile-time settings reach (`test4configs` runs those builds, but this measures one), and a few conditions the arithmetic makes impossible.

```
> make timing
```

times `ratpoints` on some curve with height bound 400 000. This maybe useful for comparisons.

Finally,

```
> make test
```

runs `test1`, `test1once`, `test1many`, `testdegrees`, `test2`, `test3`, `test4`, `testapi` and `timing`, and fails at the end if any of them failed, so that a script can tell whether the build passed. It takes about a minute. The two large-height tests are not part of it; they are meant for judging a change to the sieve rather than for checking that the build works.

4.4 Install

If you like, you can install the library, executable and header file on your system.

```
> sudo make install
```

The executable is copied to `/usr/local/bin/`, the library to `/usr/local/lib/`, and the header file to `/usr/local/include/`. You can change the `/usr/local` prefix via

```
> make install INSTALL_DIR=...
```

4.5 Debugging

In case you found a bug and would like to find out where it comes from, or if you just want to see exactly what the program is doing, you can do

```
> make debug
```

in the working directory. This will build an executable `ratpoints-debug`, which, when run, will dump loads of output on the screen, so it is best to send the output to a file, which you can then study at leisure.

4.6 Cleaning up

In order to get rid of the temporary files, do

```
> make clean
```

To remove everything except the files from the archive, use

```
> make distclean
```

4.7 Requirements

You need a C compiler (like `gcc`, which is the default specified for the `CC` variable in `Makefile`; if necessary, it can be changed there). Since version 3.0.0, the code assumes that `long` has 64 bits and refuses to compile otherwise; that is the case on every 64-bit Unix-like system (but not under the Windows ABI, where `long` has 32 bits even on a 64-bit machine). Version 2.2.4 is the last one that accommodates a 32-bit `long`.

In addition to the standard libraries, the program also requires the `gmp` (GNU multi-precision) library [\[gmp\]](#).

4.8 List of files

The archive contains the following files.

- `Makefile`
- `ratpoints.h` — the header file for programs using `ratpoints`
- `rp-private.h`, `primes.h` — header files used internally
- `gen_find_points.h.c`, `gen_init_sieve.h.c` — short programs that write additional header files depending on the system configuration
- `sift.c`, `init.c`, `sturm.c`, `find_points.c` — the source code for the `ratpoints` library
- `main.c`, `rpctest.c`, `rpapi.c` — the source code for the `ratpoints`, `rpctest` and `rpapi` executables, respectively
- `testdata.h`, `testdata-many.h`, `testdata-high-many.h`, `testdata-degrees.h` — the curves used by the test runs
- `testbase`, `testbase-many`, `testbase-high-many`, `testbase-degrees`, `testbase2`, `testbase3`, `testbase4`, `testbase-api` — the output they are expected to produce
- `verify-test4.py` — the brute-force check of `testbase4` (needs Python 3)
- `test3.sh`, `test4.sh` — the scripts that make `test3` and make `test4` run; they have to be executable, as have the next three
- `test4-configs.sh`, `coverage.sh` — what make `test4configs` and make `coverage` run
- `tune.sh` — what make `tune` runs to measure the machine-dependent constants (Section 8)
- `bench_init.c`, `bench_check.c` — benchmarks for two parts that are timed on their own: building the sieving tables, and the exact test. Neither is part of the library; the second is how to remeasure the constants of Section 8.2
- `ratpoints-doc-3.0.pdf` — this documentation file
- `gpl-2.0.txt` — the GNU license that applies to this program.

5 How to use `ratpoints`

5.1 Basic operation

Let

$$C : y^2 = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

be your curve, and let H be the bound for the denominator and absolute value of the numerator of the x -coordinate of the points you want to find. The command to search for the points is then

```
> ratpoints 'a_0 a_1 ... a_{n-1} a_n' H
```

The first argument to `ratpoints` is the list of the coefficients, which have to be integers, are separated by spaces, and are listed starting with the constant term. There is no bound on the size of the coefficients; they are read as multi-precision integers.

The degree n of the polynomial is limited to `RATPOINTS_MAX_DEGREE` (defined in `ratpoints.h`), which is set to 100 by default.

The program will give an error message when the polynomial (considered as a binary form of the smallest even degree $\geq n$) is not squarefree (e.g., if you specify two leading zero coefficients).

The following subsections discuss how to modify the standard behavior. This is achieved by adding options after the two required arguments. Some of the options have arguments, others don't; the ordering of the options and option-argument pairs is largely arbitrary. The exceptions are options with opposite effect, where the last one given counts, and the specification of the search intervals, which must be given in increasing order.

5.2 Early abort

By default, `ratpoints` will search the whole region that you specify and print all the points it finds. If you just want to find *one* point (for example when dealing with 2-covering curves of elliptic curves), you can tell the program to quit after it has found one point via the `-1` option, e.g.,

```
> ratpoints '-18 116 48 -12 30' 60000000 -1
```

If you don't want to see points at infinity, specify the `-i` option. The two options can be combined: `-i -1` will stop the program after it has found one finite point.

5.3 Changing the output

By default, `ratpoints` prints some general information before and after the points, which are given in the form $(x : y : z)$, one per line. Here, $(x : y : z)$ are the coordinates of the point considered as a point in the $(1, \lceil n/2 \rceil, 1)$ -weighted projective plane, which is the natural ambient space for the curve. The coordinates are integers with x and z coprime and $z > 0$, or $z = 0$ and $x = 1$ (for points at infinity).

There are various ways to change this behavior.

- To suppress all output except the points, use `-q` (for *quiet*).
- To add some more output explaining what the program is doing, use `-v` (for *verbose*). This has no effect if the `-q` option is also given.
- To suppress printing of the points, use `-z`.
- If you only want to list the x -coordinates of point pairs rather than individual points, use `-y`.
- There are four options that influence the format in which the points are printed:
`-f format -fs string-before -fm string-between -fe string-after`

The arguments to all of them are strings; `"\n"`, `"\t"`, `"\\"` and `"\%"` are recognized and do what you expect. The *format* string can contain markers `%x`, `%y`, `%z` that will be replaced with the x , y , and z -coordinate of the point, respectively. The defaults are empty for *string-before*, *string-between* and *string-after*, and `"(%x : %y : %z)\n"` for *format* when `-y` is not specified; otherwise `"(%x : %z)\n"`.

The effect is as follows. Before any point is printed, *string-before* is output. Then every point is printed according to *format*. Between any two points, *string-between* is output, and after the last point, *string-after* is output.

As an example, consider the effect of

```
-f "[%x,%y,%z]" -fs "{" -fm "," -fe "}"\n
```

5.4 Restricting the search domain

There are two ways to restrict the domain of the search. The first is to restrict the range of denominators considered. This is done via

`-dl dmin -du dmax .`

For example, to only look for integral points, you can say `-du 1`. By default, the lower limit is 1 and the upper limit is H .

The other way is to restrict the search to a union of (closed) intervals. If you only want to search for points in

$$[l_1, u_1] \cup [l_2, u_2] \cup \cdots \cup [l_k, u_k],$$

you can specify this in the form

`-l l1 -u u1 -l l2 -u u2 ... -l lk -u uk .`

The first `-l` option is optional; l_1 defaults to $-\infty$. Similarly the last `-u` option is optional, and u_k defaults to $+\infty$. The number k of intervals is bounded by `RATPOINTS_MAX_DEGREE` (usually, 100).

5.5 Setting the parameters for the sieve

It is possible to change the number of primes that are used in the various stages of the algorithm. This is done by the following options.

`-p M -N N -n n`

This sets the number of (odd) primes that are considered for the sieve to M , the number of primes that are actually used for the sieve to N , and the number of primes used in the first sieving stage to n . Giving M also fixes it: without `-p` the program considers `RATPOINTS_DEFAULT_NUM_PRIMES` of them to begin with and looks further only for a curve that needs it (Section 8.1), whereas with `-p` it looks at exactly M . The program will, if necessary, reduce M , N , and n (in this order) to ensure that $n \leq N \leq M \leq \text{RATPOINTS_NUM_PRIMES}$. The latter is 53 by default (the number of odd primes $< 2^8$), and 30 or 96 if `PRIME_SIZE` is set to 7 or 9, etc.

M has a price in memory, because the sieving tables for a prime p take p rows of p bits and so grow with p^2 : the primes below 128 need about 5 MB, those below 1024 about 1.7 GB (with 256-bit registers; twice that with 512-bit registers). Only the primes that M actually asks for are reserved, so raising `PRIME_SIZE` in order to have larger primes available costs nothing as long as M is left alone.

If `-n` or `-N` is left out, the corresponding number is chosen from the curve, as described in Section 8.1. The two constants that govern that choice can themselves be set:

`-r x -R e`

sets the target number of numerators surviving the first sieving stage per 64-bit word to x (a real number; this is what decides n), and the number of primes added to n to obtain N to e . The mnemonic is that `-r` and `-R` are to the automatic choice what `-n` and `-N` are to the explicit one. Both are machine-dependent; the compiled-in defaults (determined on the author's machine as of September 2026) are `RATPOINTS_SURVIVORS_PER_WORD` and `RATPOINTS_SP2_EXTRA` in `ratpoints.h`.

Three further options control the same choice, allowing to switch off or measure certain aspects of the prime-selecting stage:

`-U u -C c -A a`

The first sets the length of run, in 64-bit words of numerators, at which a second-stage prime just pays for setting itself up; `-R` is then the number of extra primes for a run much longer than that, and a shorter run gets proportionately fewer (Section 8.1); `-U 0` makes `-R` a flat offset. The second sets what building one row of a sieving table costs, in units of what one first-stage prime costs per word, which is what lets the program prefer a smaller prime to a larger one of the same quality;

-C 0 ranks the primes by quality alone. The third says how much of the choice the program should revise while it runs (Section 8.3): $\alpha = 0$ nothing, $\alpha = 1$, the default, the number of primes in the third stage, and $\alpha = 2$ the number in the second stage as well. It revises nothing it did not choose in the first place, so -n or -N has the same effect as -A 0.

A third sieving stage runs after those two, on the numerators that survive them. It tests each of them against further primes one at a time, computing the position in the table of admissible residues rather than reading a bit out of a sieving table, so it needs no table and can use primes for which building one would be out of the question. How many primes it uses is chosen from the curve, and

-P k

overrides that with exactly k of them; -P 0 switches the stage off, and

-Q q

sets instead the machine constant the choice turns on, namely what carrying one more prime through the denominators costs as a fraction of one exact test. It is to -P what -r is to -n. The choice weighs what a prime removes, which is a fraction of the survivors, against what it costs, which is a little work per survivor and a little more per denominator, so it depends on how many survivors a denominator has left by then — many for a curve with many rational points, few for a random one, and more at a large height bound than at a small one. On a random curve at a height bound of 16383 the stage does nothing at all, which is as it should be. Section 8.2 has the rule; the two machine constants in it are RATPOINTS_SP3_PER_SURVIVOR and RATPOINTS_SP3_PER_DENOM in ratpoints.h.

Both of those constants are fractions of what one exact test costs, which depends on the curve: the test evaluates a binary form of the given degree at one pair of integers and takes an integer square root, so it costs more at a high degree, and more again when the coefficients are large enough to push the arithmetic into another limb. The program estimates this cost from the degree, the largest coefficient and the height bound, and

-W w

overrides the estimate with w rdtsc cycles. An rdtsc cycle is one tick of the processor's time-stamp counter, which the x86 instruction rdtsc reads; on current processors that counter advances at a fixed rate close to the nominal clock rate, whatever frequency the core is running at, so it measures time — in units of a fraction of a nanosecond — rather than the core's own cycles. It is the unit in which the program measures itself: make bench_check reports it, and RATPOINTS_CHECK_REFERENCE in ratpoints.h is expressed in it. -W 306 assumes that every curve costs what a degree-6 curve with small coefficients costs (on the author's machine). Section 8.2 has the formula, and -v prints what it gives for the curve in hand.

There are two more parameters that can be set.

-F D

sets the maximal number of 'forbidden divisors'. If the degree is even, the denominator of any x-coordinate of a rational point is restricted at certain primes p: it cannot be divisible by p when the leading coefficient is a non-square mod p, and when p divides the leading coefficient its valuation at p is restricted instead — the program works out which valuations can occur at all (Section 7.5.3), which typically rules out a power of p, or one particular valuation, or p itself. Each such prime counts as one forbidden divisor. The program constructs a list of them against which to check the denominators, taking the primes up to the square root of the height bound, and this option caps the length of the list; the default is 64, which the list reaches at a height bound of about half a million. A prime in the list costs a few instructions per 64 denominators. With -v the program prints what it found.

The other option is

`-S S` or `-s`.

In the first form, it sets the number of refinement (interval halving) steps in the isolation of the real components to S . This part of the program computes a Sturm sequence for the polynomial and uses it in order to find a union of intervals that contains the intervals of positivity of the polynomial. This is done by a successive subdivision of the interval $]-\infty, +\infty[$. The number S sets the recursion depth. S can be omitted; then it is given a default value. The option `-s` skips the Sturm sequence computation completely. This also has the effect of removing most of the check for squarefreeness.

5.6 Switching off optimizations

The options `-k` and `-j` can be used to prevent the program from reversing the polynomial, which it usually does if this will lead to faster operation (`-k`), and to prevent the program from using the Jacobi symbol test on the denominators (`-j`). This last test extends the ‘forbidden divisors’ method described in Subsection 5.5 above by computing the Jacobi symbol (l/d) , where l is the leading coefficient and d is the odd and coprime-to- l part of the denominator. If the symbol is -1 , the denominator need not be considered. The use of these options may be questionable, unless you want to see how much performance is gained by using these optimizations.

5.7 Switching off exact testing of points

The option `-x` will prevent the program from checking the potential points that survive the sieve whether they really give rise to rational points. It effectively sets the `-y` option, since the y -coordinates are not computed. This implies that the x -coordinates that are output may not actually correspond to rational points.

5.8 Overriding previous options

The options `-I`, `-Y`, `-Z`, `-S`, `-K`, `-J`, `-X` can be used to cancel the effect of the corresponding lower-case option (and thereby restore the default behavior), when this occurs earlier in the list of options. This may be useful if you want to set a default behavior that is different from what the program does out-of-the-box (e.g., in a shell script), but want the caller to be able to override this change.

6 How to use the library

It is possible to use the `ratpoints` machinery from within your own programs.

The library `libratpoints.a` provides the following functions.

```
long find_points(ratpoints_args*,
                 int proc(long, long, const mpz_t, void*, int*),
                 void*);

void find_points_init(ratpoints_args*);

long find_points_work(ratpoints_args*,
                     int proc(long, long, const mpz_t, void*, int*),
                     void*);

void find_points_clear(ratpoints_args*);
```

The passing of arguments to these functions is via the `ratpoints_args` structure, which is defined as follows.

```
typedef struct { mpz_t *cof; long degree; long height;
                ratpoints_interval *domain; long num_inter;
                long b_low; long b_high; long sp1; long sp2; long sp3;
                double survivors_per_word; long sp2_extra; double sp2_u0;
                double cost_table; long adapt;
                long sp3_extra; double sp3_per_denom; double check_cost;
                long array_size;
                long sturm; long num_primes; long max_forbidden;
                unsigned int flags;
                long sp1_used; long sp2_used; long sp3_used; ...}
    ratpoints_args;
```

The dots at the end stand for additional fields that are used internally and are not of interest here. All fields up to and including `flags` are inputs, and they come back from a call as they went in, with three exceptions that are described below: the coefficient array and degree, which on return describe the polynomial the program worked with; `num_inter` and `domain`, which on return describe the region it searched; and the flag bits that report on the run. (Up to version 2.2.4 the program also wrote the values it chose or normalised into the other input fields, so that a caller who fills the structure once and then loops over curves had to reset them before every call; this is no longer necessary.) Every input field has to be set before the call; a negative value asks for the compiled-in default, or for the value to be chosen from the curve, as described below. The three `_used` fields are outputs; see below.

When calling `find_points`, the `cof` field must point to an array of size `degree + 1` of properly initialized gmp integers; these are the coefficients of the polynomial. The program can alter the values of the integers in this array: it may reverse the polynomial, and it drops a zero constant term on that occasion, in which case `degree` is lowered by one so that `cof` and `degree` still describe the polynomial the program worked with (the `RATPOINTS_REVERSED` flag says when this has happened). Leading zero coefficients are dropped in the same way, lowering `degree`, without a flag. To prevent the reversal, set the `RATPOINTS_NO_REVERSE` bit in `flags`; this may result in a loss of performance, though.

`height` gives the height bound. It is an error for the degree or the height bound to be nonpositive; in this case the function returns the value `RATPOINTS_BAD_ARGS`.

The field `domain` must contain a pointer to an array of `ratpoints_interval` structures of length at least `num_inter` plus `degree`. This array gives (in its first `num_inter` entries) the intervals for the search region. The type `ratpoints_interval` is just

```
typedef struct {double low; double up;} ratpoints_interval;
```

the meaning of this should be clear. In `ratpoints`, this is set via the `-l` and `-u` options. Usually, you do not want to restrict the range of x-coordinates; then you set `num_inter = 0` (but you still have to fill `domain` with a valid pointer to at least `degree` intervals; a null pointer makes the call return `RATPOINTS_BAD_ARGS`, since the program stores the default interval `]-height,height[` there. If `sturm` is negative, one interval is enough.) The program intersects the given intervals with the region where `f` is positive (unless `sturm` has a negative value, which may lead to a performance loss) and searches the result; on return, `num_inter` and the first `num_inter` entries of the array describe that result, which is why the array has to be longer than the input. So `num_inter` (and `domain`, if you restrict the search) must be set before every call.

`b_low` and `b_high` carry the lower and upper bounds for the denominator; if non-positive, they are set to 1 and the height bound, respectively. In `ratpoints`, these fields are set by the `-dl` and `-du` options.

`sp1` and `sp2` specify the number of primes to be used in the first sieving stage and in the two first sieving stages together, respectively. If negative, they are chosen from the curve, as described in Section 8.1; this is what the command line program does when `-n` and `-N` are absent. In `ratpoints`, these fields are set by the `-n` and `-N` options. What was actually used, chosen or not, is reported in `sp1_used` and `sp2_used`. The fields `survivors_per_word` and `sp2_extra` hold the two constants that govern that choice (options `-r` and `-R`); if negative, the values compiled into `ratpoints.h` are used. So do `sp2_u0` and `cost_table` (options `-U` and `-C`), which say how long a run has to be for a second-stage prime to pay for setting itself up and what one row of a sieving table costs; zero in either switches off the corresponding refinement. `adapt` set to zero stops the program revising these numbers as it runs (option `-A`, Section 8.3); it revises them only when it chose them, so a non-negative `sp1` or `sp2` has the same effect.

`sp3_used` reports the number of primes used in all three stages together, so that `sp3_used - sp2_used` of them belong to the third. (`sp3` itself is a working field of no interest to the caller.) What goes in is `sp3_extra`, which fixes that difference when it is non-negative (option `-P`) and otherwise leaves the number to be chosen from the curve, and `sp3_per_denom` (option `-Q`), which is the machine constant that choice turns on and takes the value compiled into `ratpoints.h` when negative. `check_cost` (option `-W`) says what one exact test costs, in `rdtsc` cycles (Section 5.5), and both of those constants are fractions of it; when negative, which is the usual case, the program estimates it from the degree, the coefficients and the height bound as described in Section 8.2. Similar statements are true for `num_primes` (option `-p`), `sturm` (options `-s`, `-S`) and `max_forbidden` (option `-F`). The field `array_size` specifies the maximal size (in bit arrays; a bit array contains 64, 128, 256 or 512 bits, depending on the configuration) of the array that is used in the first sieving stage. If non-positive, it is set to a default value. The various default values are defined at the beginning of the header file `ratpoints.h`, where also the maximal degree of the polynomial is set.

The `flags` field holds a number of bit flags.

- `RATPOINTS_NO_CHECK` — when set, do not check whether the surviving x -coordinates give rise to rational points (set by the `-x` option to `ratpoints`).
- `RATPOINTS_NO_Y` — only list x -coordinates (in the form $(x : z) \in \mathbb{P}^1(\mathbb{Q})$) instead of actual points (with a y -coordinate); this is set by the `-y` option to `ratpoints`.
- `RATPOINTS_NO_REVERSE` — when set, do not allow reversal of the polynomial (set by the `-k` option to `ratpoints`).
- `RATPOINTS_NO_JACOBI` — when set, prevent the use of the Jacobi symbol test (set by the `-j` option to `ratpoints`).
- `RATPOINTS_VERBOSE` — when set, causes the procedure to print some output on what it is doing (set by the `-v` option to `ratpoints`).

There are some other flags that are used internally. One of them might be of interest:

- `RATPOINTS_REVERSED` — when set after the function call, this indicates that the polynomial has been reversed (and the contents of the `cof` array have been modified).

The main vehicle for passing information back to the caller is the `proc` function argument together with the pointer `info`. This function

```
int proc(long x, long z, const mpz_t y, void *info, int *quit)
```

is called whenever a point was found. x , y and z are the coordinates of the point (where y is a `gmp` integer). `info` is the pointer that was passed to `find_points`; this can be used to store

information that should persist between calls to `proc`. If `*quit` is set to a non-zero value, this indicates that `find_points` should abort the point search and return immediately; otherwise the search continues. This is how the `-1` option works. The return value is taken as a weight for counting the points; usually it will be 1.

The usual framework for using `find_points` is as follows.

```
(...)
#include "ratpoints.h"

(...)

mpz_t c[RATPOINTS_MAX_DEGREE+1]; /* The coefficients of f */
ratpoints_interval domain[2*RATPOINTS_MAX_DEGREE];
        /* This contains the intervals representing the
           search region */

/*****
 * function that processes the points
 *****/

typedef struct {...} data;

int process(long x, long z, const mpz_t y, void *info0, int *quit)
{ data *info = (data *)info0;

    (...)

    return(1);
}

/*****
 * main
 *****/

int main(int argc, char *argv[])
{
    long total, n;
    ratpoints_args args;

    long degree      = 6;
    long height      = 16383;
    long sieve_primes1 = RATPOINTS_DEFAULT_SP1;
    long sieve_primes2 = RATPOINTS_DEFAULT_SP2;
    long num_primes   = RATPOINTS_DEFAULT_NUM_PRIMES;
    long max_forbidden = RATPOINTS_DEFAULT_MAX_FORBIDDEN;
    long b_low        = 1;
    long b_high        = height;
    long sturm_iter    = RATPOINTS_DEFAULT_STURM;
    long array_size    = RATPOINTS_ARRAY_SIZE;
    int no_check       = 0;
```

```

int no_y          = 0;
int no_reverse    = 0;
int no_jacobi     = 0;
int no_output     = 0;

unsigned int flags = 0;

data *info = malloc(sizeof(data));

/* initialize multi-precision integer variables */
for(n = 0; n <= degree; n++) { mpz_init(c[n]); }

(...)

{ /* set up polynomial */
    long k;
    for(k = 0; k < 7; k++) { mpz_set_si(c[k], ...); }

    args.cof          = &c[0];
    args.degree       = 6;
    args.height       = height;
    args.domain       = &domain[0];
    args.num_inter    = 0;
    args.b_low        = b_low;
    args.b_high       = b_high;
    args.sp1          = sieve_primes1;
    args.sp2          = sieve_primes2;
    args.survivors_per_word = -1.0; /* negative: the compiled-in default */
    args.sp2_extra     = -1;
    args.sp2_u0        = -1.0;
    args.cost_table    = -1.0;
    args.adapt         = -1;
    args.sp3_extra     = -1;          /* negative: chosen from the curve */
    args.sp3_per_denom = -1.0;
    args.check_cost    = -1.0;
    args.array_size    = array_size;
    args.sturm         = sturm_iter;
    args.num_primes    = num_primes;
    args.max_forbidden = max_forbidden;
    args.flags         = flags;

    info->... = ...;
    (...)

    total = find_points(&args, process, (void *)info);
    if(total == RATPOINTS_NON_SQUAREFREE)
    { ... }
    if(total == RATPOINTS_BAD_ARGS)
    { ... }

```

```

    (...)

}

/* clean up multi-precision integer variables */
for(n = 0; n <= degree; n++) {mpz_clear(c[n]); }

return(0);
}

```

If points are to be searched on many curves of the same degree, then it is slightly more efficient to use the sequence

```

args.degree = degree; /* this information is needed */
find_points_init(&args);

for( ... )
{ ...
    total = find_points_work(&args, process, (void *)info);
    ...
}

find_points_clear(&args);

```

This avoids the repeated allocation and freeing of memory. Inside the loop only `cof`, `degree` and `num_inter` (and whatever else changes from curve to curve) need to be set; the other input fields keep their values across calls. `rptest.c` with the `-O` option does exactly this, and `make testonce` checks that the points found are the same as with the fields set once per curve.

For practical examples, see `main.c` (the code that wraps `find_points` for the command line program `ratpoints`) or `rptest.c` (which runs `find_points` on some test data).

7 Implementation

7.1 Overview

Let $F(x, z)$ be the binary form of even degree corresponding to the polynomial on the right hand side of the curve equation. The basic idea is to let b run from 1 to the height bound H , and for each b , let a run from $-H$ to H , and finally, for each coprime pair (a, b) check if $F(a, b)$ is a square.

Of course, in this form, this would take a very long time. To speed up the process, we try to eliminate quickly as many pairs (a, b) as possible before the actual test. This can be done by ‘quadratic sieving’ modulo several primes: if $F(a, b)$ is a square, it certainly has to be a square mod p , and so we can rule out all (a, b) that do not satisfy this condition. Another ingredient is to represent (for a fixed b) the various values of a by bits and treat all the bits in a ‘bit array’ (of 64 bits, or of 128, 256 or 512 bits when SSE/AVX instructions are used) in parallel. For this, we organize the sieving information for each prime p into p arrays of p words each, one such array for every $b \bmod p$, such that the j th bit in this array is set if and only if $F(j, b)$ is a square mod p .

For each ‘denominator’ b , we then set up an array of words whose bits represent the range $-H \leq a \leq H$ (aligned so that $a = 0$ corresponds to the 0th bit of a word); the bits are initially set. Then for each of the sieving primes p , we perform a bit-wise *and* operation between this array and

the sieving information at p . In a first stage, this is done on the whole array; after this first stage, each remaining ('surviving') non-zero word in the array is subject to tests with more primes. If some bits are still set after this second stage of sieving, the corresponding pairs (a, b) (if coprime) are tested against a few further primes one at a time, without sieving tables (the third stage), and those that pass are then checked exactly.

In the following subsections, we discuss a number of improvements that were made.

7.2 Sorting the primes

This idea is due to **Colin Stahlke**. We do not just take the first so many primes in increasing order for the sieving, but we first compute the number of points the curve has mod p for a number of primes p and then sort the primes according to the fraction of x -coordinates that give points. We then take those primes for the sieving that have the smallest fraction of 'surviving' x -coordinates. In this way, we need fewer sieving primes to achieve a comparable reduction of point tests. Since version 3.0.0 the candidates include the odd numbers below 64 (a candidate sharing a prime with one already taken is passed over), and the ranking also takes into account what they cost to set up (Section 8.1).

7.3 Using connected components

We note that we can only have points when $F(a, b)$ is non-negative. If we can determine intervals on which $f(x) = F(x, 1)$ is negative, then we do not have to look for points in these intervals. The necessary computations can be performed exactly, by computing a Sturm sequence for f and counting the number of sign changes at various points, see [Coh, Thm. 4.1.10]. This can in particular tell us whether F is negative definite, in which case we have already proved that there are no rational points in the curve. If there are real zeros, we use a subdivision method in order to find a collection of intervals containing the projections of the connected components of the curve over $\mathbb{R}$.

7.4 Using 2-adic information

John Cremona suggested that in some cases, one can determine beforehand that all points will have odd numerators a , and so we can pack the bits more tightly by only representing odd numbers. This approach can be extended. We first find all solutions mod 64. Then for every residue class mod 64 of the denominator b , we can find the residue classes mod 64 of potential numerators a . If there are none, then we can eliminate b as a denominator altogether. If all potential a 's lie in one residue class modulo 2^k , we can restrict the sieving to such numerators ($2^k = 2, 4$ and 8 occur on the test curves), and the arrays are shorter by a factor of 2^k .

7.5 Elimination of denominators

Depending on the equation, certain denominators can be excluded. We have seen an instance of this in the previous subsection, but we can also work with odd primes.

7.5.1 Odd degree and $\pm$ monic

If the polynomial has odd degree and leading coefficient ± 1 , then the denominator has to be a square. This reduces the time complexity tremendously, from $O(H^2)$ to $O(H^{3/2})$.

7.5.2 Odd degree general

In general, when f has odd degree, the denominator has to be ‘almost’ a square: it must be a square times a (squarefree) divisor of the leading coefficient, and there are further restrictions on the parity of the valuation at p , for p dividing the leading coefficient, when this valuation is sufficiently large. This gives the same type of time complexity as in the monic case, but with a larger constant.

7.5.3 Even degree

Let l be the leading coefficient. If l is not a square, we can exclude denominators divisible by a (small) prime p such that l is a non-square mod p : for $p \mid b$ we have $F(a, b) \equiv la^n \pmod{p}$ with $p \nmid a$, which is a non-square as well. In addition, we use the necessary condition that the Jacobi symbol $(\frac{l}{b'})$ must be $+1$, where b' is the part of b coprime to $2l$. The Jacobi symbol is computed (for most denominators, at least) as a product of Legendre symbols over the prime divisors of l ; they are determined by a fast table lookup. When l has a prime factor above 1021, or more odd prime factors than the tables allow for (sixteen, or 8 KB), or the denominators exceed 2^{32} , the computation falls back on a binary algorithm.

When p divides l , we can instead consider p -adic valuations. Write $v_p(b) = m \geq 1$ and let w_j be the valuation at p of the coefficient of b^j in F , so that the term with b^j has valuation $w_j + jm$. If one term has strictly smaller valuation than all the others, it determines $v_p(F(a, b))$ and the residue of $F(a, b)/p^v \pmod{p}$. If that valuation is odd, no denominator with $v_p(b) = m$ can occur. If it is even and j is even, then a^{n-j} and $(b/p^m)^j$ are both squares mod p , so none can occur when the unit part of that coefficient is a non-square mod p . For large m the leading term wins by itself, so when $v_p(l)$ is odd, or even with a non-square unit part, every sufficiently large valuation is excluded: the denominator cannot be divisible by a certain power of p . For example, when $m = 1$, then p itself is excluded as a divisor when p divides the next coefficient, and p^2 is excluded otherwise.

While computing the sieving information, we make a list of these primes, each with the set of valuations it excludes, which we then use to eliminate denominators: a prime that is excluded altogether is tested for with bit arrays in the manner of the sieve, the others by computing the valuation of the denominator. The bit arrays cover every excluded prime up to the square root of the height bound (but at most up to 1021), building the patterns for the primes beyond the compiled sieving primes as needed. Note that past the square root a denominator can have at most one excluded prime left, and the Jacobi symbol sees that one.

7.6 Reversing the polynomial

The exclusion of denominators described above is more or less effective, depending on the situation; it is the better the earlier the case was described. Since the height of the points does not change under the transformation $(x, y) \rightarrow (1/x, y/x^{\lceil n/2 \rceil})$, we can as well search on the reversed polynomial $F(z, x)$. This will result in a speed-up if the reversed polynomial belongs to a ‘better’ class than the original.

7.7 Some general remarks

Much of what was learnt about where the time goes on a current processor while this version was developed ran against expectation.

Integer division is slow, and the earlier versions avoided it wherever they reasonably could. However, the subtraction chains written to avoid divisions turned out to not be that fast either, mostly

because they lead to many (costly) mispredicted branches (see also below); they have been replaced by reductions using a multiplication with a precomputed reciprocal.

What does cost is memory traffic and control flow. The innermost loop of the first sieving stage loads a row of a sieving table for each modulus and ANDs it into the registers, and it runs at the rate at which the first-level cache delivers those rows: an AND costs about one cycle as long as the rows of all first-stage moduli together fit into that cache (some 25 KB), and more for every kilobyte beyond. This is why the composite sieving moduli are capped at 64 although larger ones would remove more numerators per AND, why every row is stored `RBA_PACK` times over so that each load is aligned, and why nothing should be added to the structure this loop reads (see Appendix A).

A mispredicted branch costs fifteen to twenty cycles, the time of as many ANDs, and a branch that depends on the data mispredicts often. This effect was particularly pronounced in previous versions at small height bounds. The relevant parts (e.g., the Jacobi symbol, the gcd, locating the surviving bits) now avoid data-dependent branches, resulting in a reduction of the number of mispredicted branches to between 3% and 40% of what version 2.2.4 had.

For more details and further lessons, see Appendix A.

8 Fine-tuning the parameters

Most of what the sieve does is decided from the curve before a single numerator is sieved: which moduli to sieve with and how many in each of the three stages, how long the run is going to be, and what one exact test will cost. The rules weigh running-time costs against each other, so they rest on a few constants that describe the machine rather than the curve. Reasonable values (based on tests performed on the author’s machine) are compiled into `ratpoints.h`. `make tune` measures better ones (Section 8.4), and each of them can be set on the command line for a single run. This section says what the rules are and what the constants mean; Section 7 says how the sieve works, and Appendix A records how the rules were measured and where each constant came from.

8.1 How the moduli of the first two stages are chosen

While collecting the sieving information, the program computes for each modulus m — all odd numbers below 63 and the primes in the range considered — the density r_m , the proportion of residues a modulo m for which $f(a)$ is a square. The expected proportion of numerators surviving a set of (pairwise coprime) moduli is the product of their densities, so a modulus with a small density gives more information.

The moduli are not ranked by density alone, because they do not cost the same. A modulus m needs a sieving table of m rows, built once for each class of denominators modulo m that turns up — at most m times, however long the run — and an entry per denominator in the list of reduced denominators; per word of numerators both fall like $1/U$, where U is the number of words the run is going to sweep. The program computes U and the number of denominators from the search region before it looks at a modulus, and ranks the moduli by

$$\frac{\text{cost}(m)}{-\log r_m},$$

the cost per word divided by the information. At equal quality a smaller modulus is preferred, and at a small height bound strongly so. `RATPOINTS.COST.TABLE` (option `-C`) is what building one row of a table costs, in units of one first-stage AND per word; `-C 0` ranks by density alone.

The first stage is filled from the top of that ranking for as long as a modulus pays: it is taken while the survivors per word it removes, weighted by what a survivor costs in the stages after it, exceed `RATPOINTS.SURVIVORS.PER.WORD` (option `-r`) times what the modulus costs per word — one AND,

plus its tables, its per-denominator entries and the fetch of its row, spread over the run. That threshold is the ratio of two machine costs and the most important constant of all. With it the first stage stops at a few thousandths of a surviving bit per word whatever the curve. The second stage then gets

$$sp2 = sp1 + \frac{RATPOINTS_SP2_EXTRA}{1 + RATPOINTS_SP2_U0/U}$$

further primes, capped by the number that carry any information at all. `RATPOINTS_SP2_EXTRA` (-R) is the number an arbitrarily long run wants, and `RATPOINTS_SP2_U0` (-U) is the length of run at which setting a prime up costs as much as sieving with it, so that a short run gets proportionately fewer; `-U 0` makes the offset flat. The second stage has a ranking of its own, since a prime there is applied only to the bit arrays that survived the first stage, and its table therefore weighs some three times more heavily against it.

A curve with very many rational points tends to make f a square modulo every residue for the smallest primes. Those primes say nothing and are discarded, and the thirty primes the program looks at by default may not leave enough for the rule to have what it wants; the program then looks at further primes, one at a time, until they suffice or the table ends. Every prime added can only lower `sp1` and raise the number of informative primes, so this stops of its own accord; random curves do not trigger it. Giving `-p` explicitly switches it off: an explicit number of primes is a hard limit, as are `-n` and `-N` for the first two stages.

8.2 How many primes the third stage uses

The third stage tests the survivors of the first two stages one at a time, against further primes and without a sieving table. A prime saves the fraction $1 - r_p$ of the survivors that reach it, each of which would otherwise go to the exact test with its multi-precision arithmetic and its integer square root; it costs a multiplication and a reduction for every survivor tested, and one step per denominator to keep the denominator reduced modulo it, whether or not a survivor turns up. So a prime is worth adding while

$$S(1 - q_s - r_p) > q_d,$$

where S is the expected number of survivors a denominator still has and q_s , q_d are the two costs as fractions of one exact test — `RATPOINTS_SP3_PER_SURVIVOR` and `RATPOINTS_SP3_PER_DENOM` in `ratpoints.h`; option `-Q` sets the second. Since S falls by the factor r_p with each prime, this stops of its own accord. On a random curve at a small height bound the stage uses no prime at all; on a curve with many rational points at a large one it uses twenty or more, and looks further up the table of primes whenever none in hand pays but a prime of the density the curve has been offering would. `-P k` fixes the number at k instead, and `-P 0` switches the stage off.

S is estimated from the number of numerators a denominator has to consider, thinned by the 2-adic information, by the moduli of the first two stages, and by the proportion of numerators coprime to the denominator — that last because the test for common factors runs before this stage. This removes copies of x -coordinates that have already been dealt with, in particular those coming from rational points, which would survive every test.

What one exact test costs depends on the curve, not only on the machine. The test evaluates the binary form $F(a, b)$ of degree d by Horner's rule and takes an integer square root, so it costs a step per degree on numbers that grow to about $\log_2 |c| + d \log_2 H$ bits, plus a fixed amount for every further limb of the root; over the range from degree 3 with small coefficients to degree 14 with 400-bit ones it varies by a factor of ten. The program estimates it from the degree, the largest coefficient and the height bound before it looks at a prime — the constants are `RATPOINTS_CHECK_STEP`, `_LIMB`, `_CALL` and `_ROOT`, relative to `RATPOINTS_CHECK_REFERENCE`, the value for a degree-6 curve

with small coefficients — and divides the two fractions above by the ratio, so that a curve whose test is dearer gets more primes. Only ratios enter, so a machine on which every test costs more needs no change; `-W w` overrides the estimate with `w` rdtsc cycles (Section 5.5), and `-v` prints what the estimate gives for the curve in hand.

8.3 Correcting the choice while the program runs

The rules above use the densities alone, and they miss one thing by construction: if a/b gives a rational point, so does ka/kb for every $k \geq 1$ and therefore passes every prime test there is. A floor of survivors therefore outlives any amount of sieving, and only the test for common factors removes it. The program can see the floor: it counts the bit arrays that survive the first stage, the survivors of the second, and the candidates that pass the test for common factors, and fits

$$S(n) = \text{floor} + \text{chance} \cdot R(n)$$

to the counts, with $R(n)$ the product of the n smallest densities. It revises its choice after the first million words of numerators and then whenever twice as many have been swept. By default (`-A 1`) it revises only the third stage, where the number of survivors a denominator brings becomes a count rather than an estimate; `-A 2` reconsiders `sp2` as well, and `-A 0` switches the revision off. Nothing that was fixed with `-n`, `-N` or `-P` is revised.

8.4 Retuning for your machine

The five constants — the threshold, the offset, the run length at which a second-stage prime pays for setting itself up, what building a table row costs, and the third stage's cost per denominator — were fitted on the author's laptop by minimising the sum of the times of `make test1` and `make test1many`, which between them cover the two regimes that occur in practice, random curves and curves with many rational points. To repeat that on your machine, run

```
> make tune
```

It sweeps the five constants, in that order, over both test sets and writes the values it finds to `tuning.mk`, which the `Makefile` includes, so that a subsequent `make all` builds with them. No source file is touched, and deleting `tuning.mk` restores the values compiled into `ratpoints.h`. The file records the configuration it was measured for and is ignored, with a warning, if you later build with another register width or `PRIME_SIZE`.

`make tune` is deliberately not part of `make all`: it takes several minutes (about three for the 256-bit build on the author's machine, four for 128 bits, six for the plain 64-bit one and eight for the 512-bit code emulated through narrower operations), wants an otherwise idle machine, and must not be run under `make -j`. Timing this reliably is the hard part. The cost surface is flat — anything within a factor of two of a good threshold costs under 3% — whereas a laptop under sustained load slows by a quarter as it warms up, so a straight sweep would measure the clock rather than the settings. Each candidate is therefore timed back to back with the current settings and only the ratio is kept, the median over several rounds is used, and the current settings are themselves among the candidates, so that the ratio they measure against themselves shows how much noise is left. If that is more than 2% the run is declared inconclusive, and nothing is written unless the winner is better by more than 3%: a noisy run leaves a good default alone. Two consequences are worth knowing in advance. A run that reports that nothing beat the current settings has *succeeded*; on a build that is already tuned that is the right answer. And a configuration whose tests take long enough for the machine to throttle within a single round may report itself inconclusive however often it is run; raising `ROUNDS`, or using a machine that does not throttle, is then the remedy. The environment variables `ROUNDS` (how many times each setting is measured, default 3), `WARMUP`

(seconds of load before anything is counted, 20), MARGIN (0.97) and NOISE (0.02) adjust this, and R_VALUES, E_VALUES, U_VALUES and C_VALUES give the ladders of candidates.

```
> make tunehigh
```

measures on `make testhigh` and `make testhighmany` instead, at a height bound of $2 \cdot 10^5$, where the two sieving phases are nine tenths of the run; at the 16383 of `test1` two fifths of the time on the random curves goes into work the constants have little or no effect on. Use it if the runs that matter to you are long ones. One timing there is two minutes rather than three seconds, so it does not sweep the whole ladder: it starts from the settings in force — which is to say from what `make tune` found, since both read and write the same `tuning.mk` — and asks only whether a factor of two either way in the threshold, in the run length or in the table cost, or two more or fewer second-stage primes, is better (R_FACTORS, E_DELTAS, U_FACTORS and C_FACTORS set that neighbourhood). Expect a couple of hours at the default three rounds, and run it again if it moved a value. The intended order is `make tune` first and `make tunehigh` afterwards; only a pair of tunings at run lengths a hundred times apart can separate the offset from the run length it is scaled by.

Should you prefer to do it by hand, the constants can be given for a single run with `-r`, `-R`, `-U`, `-C` and `-Q`, which `ratpoints` and `rpctest` both accept, and hard-coded by changing the definitions in `ratpoints.h` (followed by `make distclean all test`). `rpctest -z -T` prints the CPU time of a run and nothing else, which is what `tune.sh` uses; `-h H` changes its height bound, `-m m` repeats the computation `m` times, and `testdata.h` can be replaced by a collection of your own curves. Programs using the library set the same constants through the fields of `ratpoints.args` (Section 6).

9 Change log

Version 2.0 was released January 9, 2008.

Version 2.0.1 was released July 7, 2008. It fixes a bug that prevented the `'-1'` option to work properly.

Version 2.1 was released March 9, 2009. It makes use of the SSE instructions, so that the sieve can work on 128 bits in parallel (instead of on 64 or 32). On my laptop (with an Intel Core2 processor), `make test` runs about 25% faster than before.

Version 2.1.1 was released April 14, 2009. It fixes a bug that in some cases prevented an early abort when `*quit` was set by the callback function. Thanks to **Robert Miller** for the bug report and the fix. In addition, it is now checked that `__WORDSIZE == 64` before SSE instructions are used (in `rp-private.h`), since the program then assumes that a bit array consists of two unsigned longs. In this context, `__SSE2__` has been replaced by a new macro `USE_SSE`. A further fix eliminates unnecessary copying of sieving information when SSE instructions are not used (introduced in version 2.1). This was almost always harmless, but could have resulted in memory corruption in the extreme case that all the information had to be computed for all the primes.

Version 2.1.2 was released May 27, 2009. It fixes some memory leaks. Thanks to **Robert Miller**.

Version 2.1.3 was released September 21, 2009. The library function `find_points` should now work without any bound on the degree of the polynomial. The degree bound for the `ratpoints` executable is now set to 100 by default (specified by `RATPOINTS_MAX_DEGREE`, used to be 10).

A bug in `rpctest.c` (pointed out to me by **Giovanni Mascellani** and **Randall Rathbun**) that could lead to a segmentation fault was fixed on March 10, 2011.

Version 2.2 was released January 9, 2022. The main change is that `ratpoints` can now also work with 256-bit (and, in principle, 512-bit) registers when the CPU has the relevant capabilities. Thanks to **Bill Allombert** for doing a first implementation of this. In addition, the code has been cleaned up to some extent (e.g., the variants according to register sizes are now completely dealt with through macros defined in `rp-private.h`), and some more comments have been added. There are also some further optimizations; for example, the first sieving phase now uses many of the SSE/AVX registers in parallel, and on some architectures, the code uses a faster implementation for the test whether a bit array is zero. On my current laptop, the new code (using 256-bit words) is about twice as fast as the old one (2.1.3 using 128-bit words).

Version 2.2.1 was released January 18, 2022. This fixes a small bug that was introduced in the previous version, which could lead to the program missing points at the upper end of the search interval for the numerators. Thanks to **Bill Allombert**.

Version 2.2.2 was released May 15, 2023. This fixes a small bug that led to the polynomials being reversed incorrectly when the degree is odd. Thanks to **Nicolas Mascot** and **Bill Allombert**.

Work on the code starting with version 2.2.3 was done with the help of **Claude Code** (by Anthropic), using various models (Opus 5, Fable 5, and Fable 5.1).

Version 2.2.3 was released September 6, 2026. It fixes the code path using 512-bit registers, which had never been exercised. Two problems were found. The first one would have made the program crash. The second problem was that the `TEST` macro, which is used in the innermost loop of the second phase, still used the generic fall-back version, leading to a degradation in performance; it now uses `vptestmq`, in analogy with the 256-bit version.

Since I have no CPU with AVX512F available, this was tested by compiling with `-DUSE_AVX512` but without `-mavx512f`, so that `gcc` expresses the 64-byte vectors through 256-bit operations. This exercises the whole code path. What this cannot check is the genuine 512-bit instructions. In the meantime, **Drew Sutherland** has run the two variants against each other on an AMD Ryzen 9 9950X3D (Zen 5 architecture), for a curve with many rational points and a height bound of 10^6 ; the 512-bit version was faster by 13 to 14%.

Version 2.2.4 was released September 19, 2026. It fixes bugs that a review of the code and running the new test suite turned up:

- A polynomial of degree 1 — given as such, or reached from $y^2 = c_2x^2 + c_1x$ by the reversal, or from a zero leading coefficient — made the Sturm sequence code read past the end of its arrays and crash; this affects library users as well.
- When the positivity region of f did not meet the search domain, the program returned before it had looked at the points at infinity, so that `ratpoints -1000000000 0 0 1' 45` printed nothing where `-s` printed `(1 : 0 : 0)`, and after a reversal the lost point is an affine point of the given curve.
- When no residue class of the denominator modulo 16 admitted any numerator, the array of numerator patterns was read before it had been filled; by default the answer was still right, but with `-j -F 0` the program printed nondeterministic points outside the height bound after a run several thousand times longer than necessary.
- Several input fields of `ratpoints_args` were used as scratch space — the number of forbidden divisors found was written into `max_forbidden`, the intersected search intervals into `num_inter`, normalised values into `sp1`, `sp2` and others — so that a program which fills the structure once and then calls `find_points_work` for curve after curve, as the section on the library suggests, had the forbidden-divisor test switched off after the first curve without forbidden divisors (a factor of two in running time) and the first curve's positivity region intersected into the search domains of the others, which loses points. The program

no longer writes either, and the manual now says exactly which fields it does set; make `test1once` runs the test curves with the structure filled once, and make `test3` runs one or more invocations of `ratpoints` per bug against a reference.

- The library tested whether the leading coefficient is a square before it checked the coefficient pointer for NULL, so that a caller passing NULL crashed instead of getting the error code back.
- An empty coefficient string made the program report “Bug no. 1 — please report!” instead of refusing the input.
- At the very top of the range of a long, three things went wrong: a search interval that reached the last few hundred numerators below 2^{63} was dropped without a word, two computations on the position of its last bit having overflowed; the loop over every denominator never ended when the denominator bound was $2^{63} - 1$, since the increment wraps around; and the loops over the square denominators formed b^2 and $d b^2$, which overflow within $2 \cdot 10^9$ of it.
- The test on the valuation of a denominator $d b^2$ at a prime dividing the leading coefficient applied `abs`, the `int` function, to a long, so that for denominators above 2^{31} it judged the low 32 bits only and could exclude a denominator that carries a point: a point with denominator $3 \cdot 59827^2$ was missed.
- Not a bug, but wasted time: the loops over the square denominators started at 1 whatever the lower bound `-d1` said; they now start at the first square in the range.
- Also fixed: four left shifts of negative values (undefined in C, harmless with gcc so far), two arrays that could be declared with length zero under `-n 0 -N 0`, an overflow in the enumeration of the divisors of the leading coefficient for denominator bounds above about $4 \cdot 10^{15}$, and two constants in `ratpoints.h` that could not be overridden from the compiler command line. The library is binary compatible with 2.2.3.

Version 3.0.0 was released September 19, 2026. Its theme is that the program works out for itself what it used to be told, and the sieve has been reworked from the set-up to the exact test. Measured against version 2.2.4 on the same curves and the same laptop (cycles pinned to one core, medians of three interleaved rounds; 2.2.4 at its own defaults, which sieve with the primes below 128 where this version uses those below 256; both find the same points on every suite):

suite	2.2.4, Gcycles	3.0.0	ratio	faster by
make test1 (random curves, 16383)	8.45	2.89	0.34	2.9
make test1many (point-rich, 16383)	11.27	4.80	0.43	2.3
make testhigh (random, $2 \cdot 10^5$)	412.7	190.8	0.46	2.2
make testhighmany (point-rich, $2 \cdot 10^5$)	764.4	138.3	0.18	5.5
random curves at height 4000	3.06	0.54	0.18	5.6
random curves at height 1000	2.06	0.21	0.10	10.1
point-rich curves at height 1000	0.44	0.15	0.34	3.0
random curves at height 200	0.96	0.12	0.13	8.0

Instructions fell to between 0.16 and 0.47 of 2.2.4’s and mispredicted branches to between 0.03 and 0.40: at the small height bounds 2.2.4 was bound by its branches, and cycles fall two to three times more than instructions there. The changes, in outline; the sections named have the details, and Appendix A the measurements behind them.

- *The sieving parameters are chosen from the curve.* How many moduli each of the first two stages uses, and which ones, follows from the densities of admissible residues, from the length of the run and from what each modulus costs, with the moduli ranked by cost per information rather than by information alone (Section 8.1); `-n` and `-N` still fix the

numbers. When the small primes say nothing, as on a curve with very many rational points, the program looks at further primes of its own accord, and the default table holds the primes below 256 (PRIME_SIZE 8, formerly 7) to give it something to reach for. The rules rest on five machine constants, set with `-r`, `-R`, `-U`, `-C` and `-Q` and measured by `make tune` and `make tunehigh` (Section 8.4), which write them to `tuning.mk`.

- *A third sieving stage* tests the survivors of the two bit-array stages one at a time against further primes, without sieving tables, so that it can use primes far up the list; how many is decided per curve (Section 8.2, option `-P`) from an estimate of what one exact test costs on that curve (`-W`). While it runs, the program counts what the sieve is finding and revises the choice (Section 8.3, option `-A`).
- *Composite moduli*. Odd numbers below 64 sieve alongside the primes (the bound is given by `RATPOINTS_COMPOSITE_MAX` in `ratpoints.h`).
- *2-adic information modulo 64* instead of 16, the even denominators decided exactly through the reversed polynomial, and the bit arrays of each denominator class packed with the largest stride 2^k its numerators allow rather than by parity alone (Section 7).
- *Denominators*. The test at a prime dividing the leading coefficient, derived from the Newton polygon of F , now runs; the forbidden-divisor bit arrays cover every excluded prime up to the square root of the height bound, and the default of `-F` is 64; the Jacobi symbol is a product of table-driven Legendre symbols; the tests on $b \bmod 64$ are applied to 64 denominators at a time; the residues of b come from precomputed reciprocals, and the gcd that tests a candidate for lowest terms is branchless.
- *The sieving loops*. The first stage ANDs the 2-adic pattern in with its first modulus instead of pre-filling the bit arrays, and sieves the bit arrays left over after the last whole chunk of sixteen in legs of 8, 4, 2 and 1 registers instead of padding the range; the second stage steps over the empty bit arrays into a sentinel, reaches its table rows through a precomputed reciprocal and extracts the survivors by counting trailing zeros. The sieving tables are built four times faster, and the memory reserved for them follows the number of primes in use.
- *Build and portability*. A stamp file makes a change of `CCFLAGS1` or `PRIME_SIZE` rebuild what depends on it; `CCFLAGS128` selects `USE_AVX128`, which needs only SSE2 and is faster than the `USE_SSE` variant; the code assumes a 64-bit long and refuses to compile without one, so version 2.2.4 is the last one for 32-bit machines.
- *Library*. `ratpoints_args` has gained fields, so programs using the library must be recompiled; the input fields come back as they went in, except `num_inter` and `domain`, which describe the region searched, and `sp1_used`, `sp2_used` and `sp3_used` report what was used (Section 6).
- *Tests*. `make test` runs, besides the random curves, the point-rich ones (`test1many`), a hundred curves of degrees 3, 4, 7 and 8 (`testdegrees`), the argument structure filled once for all curves (`test1once`), one invocation per bug fixed (`test3`), a suite that exercises every branch of the code against a reference checked by brute force (`test4`) and the library interface (`testapi`), and it fails when a test fails; `testhigh`, `testhighmany`, `test4configs` and `coverage` are run separately (Section 4.3).

A Notes on the optimizations

This appendix was written by Claude Code, the AI assistant (by Anthropic) with whose help this version was developed (Section 9); it consolidates the notes kept during that work. It keeps what a reader would need in order to change the program without repeating the experiments: how the measurements were made and what they can and cannot resolve, where the time goes, why curves

with many rational points behave so differently from random ones, the cost model with the origin of each constant, what each change was worth, what was tried and did not pay, and how much of all this depends on the machine. Every figure was taken on one laptop — an Intel i7-1355U, the code pinned to one performance core, the 256-bit build, gcc 14 — and is quoted as a ratio of cycle counts, the program with the change in question over the program without it; Section A.7 says what would come out differently elsewhere.

A.1 How the measurements were made

The effects in question are mostly between one and ten per cent, and the machine is noisier than that. Under sustained load it slows by about a quarter as it warms up, over tens of seconds, so wall-clock times taken minutes apart mean nothing; core cycles counted by `perf` on one pinned core do better, but the drift affects them too. Every comparison was therefore made in pairs: the candidate and the baseline run back to back, interleaved, for several rounds; only the ratio of each round is kept, and the median of the per-round ratios is quoted, not the ratio of the medians (the two differ, because the drift is not symmetric between the two positions). The baseline is run against itself as one of the candidates, and its ratio measures the noise that is left; a run in which that exceeds two per cent is not believed. `tune.sh` does all of this and refuses to write anything a noisy run suggests.

Two floors remain below which whole-program timings cannot decide anything. Where the compiler places a hot loop is worth up to ten per cent on this machine: the same source rebuilt with `-falign-loops=32` and `64` gave ratios of `0.98` and `1.00` where the default placement gave `1.09`. Whole-program comparisons were therefore made at three placements of the code, and a change under about two per cent at a height bound of $2 \cdot 10^5$ — where half the run is the innermost loop of the first stage, which is exactly what those flags move — cannot be settled that way at all. The other method, putting both variants in one binary behind a switch read once at start-up, removes the placement effect entirely and is the better evidence; but it is blind to a data-layout cost. A change that grows a structure read in the innermost loop by eight bytes shows nothing within one binary and gives back half of a 2.8% gain across builds. Both methods were used when a change touched such a structure.

Instruction counts are exact, repeatable to 10^{-8} , and the right tool for attributing a set-up cost to a function; as a proxy for time they misled three times. `-funswitch-loops` removed one to three per cent of the instructions and gained nothing in cycles at any placement; hoisting the per-curve constants of the sieve entries out of the per-denominator loop removed 2.4% of the instructions and 0.1% of the cycles, the loop being bound by its chain of dependent loads; and composite moduli above 64 removed 22% of the instructions of `make testhigh` while the cycles fell less than with the bound in place and `make testhighmany` lost 7%, because the rows of a large modulus do not stay in the first-level cache. Cycles decide; instructions explain.

The test suites see different things. At the height bound 16383 of `make test1` and `test1many` two fifths of the time on the random curves is not sieving at all but choosing the primes, building the tables and setting up each denominator; at the $2 \cdot 10^5$ of `testhigh` and `testhighmany` the two sieving stages are nine tenths of the run. A change to the sieving loops is therefore judged on the latter, a change to the set-up on the former or at height bounds of 200 to 4000, and the two regimes of Section A.3 are always reported apart, since they often disagree and the disagreement is usually the finding. One more thing a correctness suite cannot see: a change that removes an initialisation reproduces every reference output, because the values it stops writing were about to be overwritten — and a debug or instrumented build may still read them. The `-DDEBUG` and the `-DRP_PHASE_TIMING` builds are run under `valgrind` after such a change.

A.2 Where the time goes

The sieving tables are built once for each pair of a modulus and a class of denominators modulo it and kept, so their cost is bounded by the moduli and does not grow with the height bound; everything else grows at least linearly with it. The shares therefore depend on the height bound above all. On the random test curves:

	height 16383	height $2 \cdot 10^5$
the two sieving stages, with the exact checks	57%	90%
the sieving tables	4%	0.3%
the set-up per denominator	8.5%	1.3%
outside the sieve altogether: examining and choosing the moduli	16%	3%

(the rest is the work of `sift()` around the stages). Within the sieve the first stage is the larger part, of the order of three fifths of a long run on random curves and three quarters on point-rich ones; the second stage with the exact checks is most of the rest. The exact check itself is under one per cent of every suite, and the integer divider is busy for less than 0.6% of the cycles: neither is where the time goes (Section 7.7). At a height bound of 200 the shape is different again: examining the thirty primes is a quarter of the instructions, and the sieving tables are a twentieth of the run — they would be a fifth if the first-stage rule did not price them (Section 8.1).

The first stage’s inner loop is close to what the machine can do: sixteen fused load-and-AND instructions per modulus per chunk and about eight instructions of overhead, at 1.8 instructions per cycle, bound by the load throughput and the capacity of the first-level cache rather than by arithmetic. Its working set at the parameters a random curve uses is some 50 KB at 256 bits, which is about the size of that cache on the development machine; the marginal cost of a first-stage prime is flat while the set fits and rises past it. An AND costs one cycle while the rows of the first-stage moduli together occupy under some 25 KB, and about 0.8% more for every further KB. The mispredicted branches, of which version 2.2.4 had many, scale with the survivors and not with the ANDs.

A.3 The two regimes

Almost every tuning question in the program turns on the proportion of numerators that survive the first sieving stage, and two populations of curves behave very differently in it. On a random curve about one numerator in 10^4 survives, and the rule of Section 8.1 reaches its target with ten or twelve moduli. On a curve with very many rational points f is a square modulo every residue for the smallest primes, so those primes carry no information and are discarded, and the rule needs twenty or more primes; among the thirty primes the program examines first, about one such curve in five does not find enough. That is why the program looks at further primes when it needs to, and why the table holds the primes below 256: it is worth a factor of 1.3 to 2 on those curves at a large height bound and costs nothing on any other. Larger primes are what these curves need: the proportion of admissible residues tends to one half as the prime grows however many points the curve has, while for the small primes it is close to one.

The second thing that separates the regimes is a floor of survivors that no sieving removes: if a/b is a rational point then so is ka/kb for every k , with the same value of f , and it passes every prime test. There are about H/b of them for each point of denominator b , so they grow with the height bound while the survivors that die by chance grow with its square; only the test for common factors removes them, which is why that test runs before the third stage. The floor is what the densities alone cannot predict and what the correction of Section 8.3 measures, and it is why the number of survivors a denominator brings to the third stage ranges over a factor of 260 between the suites — 0.035 on the random curves at 16383, 9.2 on the point-rich ones at $2 \cdot 10^5$ — so that no fixed number of third-stage primes can serve and the rule there is a stopping rule.

The two regimes want the same constants, which was not obvious and was measured twice: `make tunehigh`, which tunes at $2 \cdot 10^5$, keeps every value that `make tune` finds at 16383, all within 0.7% except the threshold, which is sharper — doubling it costs 5.6% at $2 \cdot 10^5$, and two and a half times its value costs 8.6% at 16383. One threshold also serves every height bound from 200 to $2 \cdot 10^5$. That depends on the estimates behind the rules — the words the run will sweep, the denominators it will visit, the bits set per word — being right together (Section A.4): with one of them off, every height bound wants a threshold of its own.

A.4 The cost model and where the constants came from

The rules of Section 8 compare costs, and every cost in `ratpoints.h` is in one unit: what one first-stage AND costs per 64-bit word of numerators, a quarter of one AND on a 256-bit array, about 0.26 core cycles on the development machine. The constants that are not fitted were measured with the instrumented build (`-DRP_PHASE_TIMING -DRP_PHASE_COUNTS`), dividing each part's cycles by the number of times it ran. The instrumentation is not free — the counters alone can be nearly a fifth of an instrumented run — so shares are taken from a timing-only build. Table 1 lists the constants: what each one is, how its value was found, and how much a different value costs.

Two properties of the model are worth more than any single value. First, the rules sit at their own optimum, so a constant off by a factor of two costs under three per cent, and a better estimate of one moves the boundary by about one prime, which is by definition worth about what it costs; that is why the per-curve estimate of the exact check's cost changes no running time by half a per cent and is there so that the constants mean what they say. Second, the estimates the rules feed on — the words the run will sweep, the denominators it will visit, the bits set per word, the density of a modulus — and the fitted constants form one system: the constants are fitted with the estimates in place and absorb whatever error the estimates have, so a more accurate estimate on its own makes the program slower, and has to be judged after a refit of the constants (`make tune`), not before. As they stand, the predicted words per denominator are within 1.5% of the count at every height bound from 200 to $2 \cdot 10^5$, and the automatic choice of the first-stage moduli beats or ties the best fixed `-n` in every one of eighteen combinations of suite, height bound and modulus cap. A finer point of the same kind: three conventions for the density of a prime and of its powers were measured, and the mixed one — the prime approximated, the power exact — is the best on every suite. A fitted model absorbs its approximations, and “more exact” can be worse.

A.5 What each change was worth

Table 2 gives, for each change, the cycles with it over the cycles without it on the four suites — random and point-rich curves at the height bound 16383 (`make test1` and `test1many`) and at $2 \cdot 10^5$ (`testhigh` and `testhighmany`) — and, where it matters, at other height bounds or on the suite of curves of degrees 3, 4, 7 and 8 (`make testdegrees`). Each was measured as described above, against the program as it stood when the change was made, so the rows do not multiply out to the totals of Section 9. A dash is a change within the noise; an empty cell was not measured.

The degree suite gained most from the changes to the model because it is the one set none of the constants had been fitted on: rules carry over where fitted constants do not. The largest single gain on the random curves at 16383 came from the denominator side, where the Jacobi symbol computed by a binary algorithm for every denominator had been 12% of the run and a third of its mispredicted branches; the largest at $2 \cdot 10^5$ from the stride, which sweeps 15% fewer bit arrays.

constant (RATPOINTS...)	value	what it is; how the value was found	how flat
SURVIVORS_PER_WORD	0.003	the threshold at which the first stage stops taking moduli; fitted by make tune at 16383	0.0045 costs 2.7% and 0.0075 costs 8.6%; 0.0015 to 0.002 cost nothing; the same at $2 \cdot 10^5$
SP2_EXTRA, SP2_U0	11, $1.6 \cdot 10^6$	the second-stage offset of an arbitrarily long run, and the run length at which a prime's set-up costs as much as its sieving; fitted to four measured optima — 3, 5, 9 and 12 second-stage primes at $U = 5.9 \cdot 10^5, 5.1 \cdot 10^6, 8.2 \cdot 10^7$ and $7.1 \cdot 10^8$ words — which the pair reproduces as 3, 8, 11, 11	a flat offset ($-U \ 0$) costs about 5% at one height bound or the other; a flat 5 costs 12% of testhighmany; only tune and tunehigh together separate the two
COST_TABLE	38	one row of a sieving table; measured at 18 to 30	flat from 20 to 70: 0.9515 at 38 against 1 at 0, 0.9619 at 140
COST_BP, COST_SETUP	8, 30	a denominator's entry in the list of reduced denominators, and the fill of a sieve entry per denominator; measured	8 against 24 within 0.3%
COST_CALL, COST_LINE	50, 4.5	a call of the sieve per modulus (12 to 13 cycles) and the fetch of a table row per denominator (1.2 cycles per cache line); measured	not swept singly; fitted as one group with the threshold
COST_PHASE2, COST_SURVIVOR	110, 1400	one second-stage AND, and one survivor of the second stage; the first measured at 155 (random) and 61 (point-rich) at $2 \cdot 10^5$	60, 100 and 220 within about 2% of 110
SP3_PER_SURVIVOR, SP3_PER_DENOM	0.055, 0.013	the third stage's costs per survivor tested and per denominator, as fractions of one exact check; measured: 250 to 280 cycles for a check beyond the first of a denominator, some 190 once per denominator, 3.5 per step carrying b past a prime	the per-denominator value from 0.003 to 0.05 within 1.2%; a fixed -P never beats the rule by more than 0.3%
CHECK_STEP, _LIMB, _CALL, _ROOT	29, 8, 94, 170	the exact check, in rdtsc cycles: a Horner step, a further limb, the call, a further limb of the root; from make bench_check (in the sieve the check costs $44 + 1.44$ times the microbenchmark)	only ratios between curves enter; the estimate agrees with the sieve to 6% over degrees 3 to 12
CHECK_REFERENCE	306	the same for a degree-6 curve with small coefficients, which is what -W 306 assumes for every curve	scaling every curve's value alike from 306 to 1000 moves no suite by 0.5%

TABLE 1. The constants of the cost model: their values, where they came from, and how much a different value costs.

A.6 What was tried and did not pay

- *The second stage on narrower registers.* The guess was that a wider array holds more bits and so takes more primes to clear. The counters say the number of second-stage AND steps is the same to 1.5% at 64, 128, 256 and 512 bits, and a hybrid that scans at the register width and sieves the surviving words one at a time (which is what USE_LONG_IN_PHASE_2

change	height 16383		height $2 \cdot 10^5$		
	random	rich	random	rich	elsewhere
sp1, sp2 chosen from the survival rate; the tight scan over the empty arrays	0.86	0.64			
further primes for starved curves; PRIME.SIZE 8	–	–	–	–	0.50 to 0.75 on the curves it applies to
survivors located by counting trailing zeros	0.99	0.95	0.98	0.98	
the table set-up: the residue test's branch becomes a shift, four rows at a time	0.87	0.98	–	–	the tables $3.7\times$ faster
the third sieving stage	–	0.98	0.98	0.93	
reductions by a multiply-high instead of a subtraction chain	0.97	0.97	0.98	0.98	
the run length known in advance: cost-aware ranking, the offset scaled, the survivors counted, the check's cost estimated	0.94	–	–	0.88	0.86 on the degree suite
the first stage writes the arrays: no fill pass	0.97	–	0.94	0.97	0.95 on the degree suite at $2 \cdot 10^5$
the valuation test at primes dividing the leading coefficient	0.92	0.98	0.91	0.98	0.95 on the degree suite
the scan into a sentinel; the second stage's row by the reciprocal	0.97	0.97	0.93–0.96	0.93–0.96	
the denominator side: Legendre-symbol tables, 64 denominators a word, reciprocals, a branchless gcd, forbidden divisors up to $\sqrt{H}$, the third stage set up on demand	0.76	0.96	0.94	0.98	
no padding: the tail in legs of 8, 4, 2, 1 words	0.90	0.94	0.98	0.98–1.00	0.81 at 1000 and 4000, 0.92 at 200
the 2-adic information modulo 64	0.97	0.98	0.97	0.99	0.97–0.98 at 200 to 4000
the third stage looks further; even denominators at half width	–	0.99	–	–	$4.6\times$ fewer exact checks on the point-rich curves at $2 \cdot 10^5$
each denominator class packed with its 2-adic stride	0.91	0.99	0.875	0.98	0.97 at 4000, 1.035 at 1000, 1.05 at 200
composite moduli up to 64	0.92	–	0.89	0.96	nothing below a few hundred
the first-stage rule prices the tables and the downstream cost	–	–	–	–	0.76 at 200, 0.86 at 1000, 0.99 at 4000
the estimates of the run (words, denominators, bits per word, densities) made accurate, the constants refitted	0.995	0.985	0.993	0.981	1.00 to 1.03 at 200 to 4000

TABLE 2. What each change was worth: cycles with it over cycles without it, on the random and the point-rich curves at the height bounds 16383 and $2 \cdot 10^5$; a dash is a change within the noise.

selects) is a wash to 4% slower: a narrow and a wide read of the same table come from one cache line, so there was never any traffic to save.

- *Byte rotations in the table set-up.* Holding the p-bit pattern in eight shifted copies makes every row an unaligned load; but the rotation was never the expensive part — three quarters of the set-up was the loop that builds the pattern, whose branch on a quadratic-residue test mispredicted half the time — and the byte offset still steps through the same three-cycle chain. Worth 2.5% of a set-up that is a tenth of make test1, against eight scratch buffers and a byte-order assumption. The lesson that did pay: a loop carrying $x += d$; if

$x \geq p$ then $x \text{ --} p$ costs three cycles a row whatever else it does, and the remedy is to run several of them at once.

- *Sieving tables without the replication.* The tables hold each p -bit row `RBA_PACK` times over so that every bit array read is aligned; dropping that cuts their footprint by nearly that factor and is legal, since vector loads need no alignment. Priced in isolation by shifting every table one word, the split loads cost 28% of the first stage at 256 bits, against a cache cliff worth at most 17% and only for curves that reach eighteen or more first-stage primes; worse with 512-bit registers, where seven offsets in eight split a line. Ordering the tables by prime, proposed for the same reason, is the case in effect already, since they are built lazily in the order of use; and the TLB is not a constraint.
- *Testing several bit arrays at once in the scan* pays only when the proportion of surviving arrays varies widely, as it does when the number of first-stage primes is fixed; with the rate pinned at a few per cent by the rule, a group containing a survivor wastes its whole OR, and a group of one wins. Per-chunk survivor flags computed in the first stage were prototyped three ways: the branchy form mispredicted, the branchless forms spilled, since all sixteen vector registers hold live arrays. The sentinel was the answer.
- *Compiler switches.* `-funswitch-loops` removes one to three per cent of the instructions and none of the cycles; profile-guided optimisation removed 6.5% of the instructions and added two to eight per cent of cycles. Both declined.
- *Composite moduli above 64*, and a footprint term in the cost model to price them: the products of two mid-sized primes save up to 45% of the first stage's ANDs on some random curves at $2 \cdot 10^5$ and 3.7% of `make testhigh`, but a row of 220 bit arrays leaves the first-level cache and the point-rich suite loses 4.5 to 7%. Telling the two apart needs a model of the row set with three or four cache constants; the bound stays. The 2-adic pattern modulo 256 would remove one admissible class in twelve more for sixteen times the set-up.
- *Reconsidering `sp2` from the counted survivors* (`-A 2`) is a second answer to a question the offset scaled by the run length already answers, and whichever is applied later wins: without the offset it costs 5% of `make testhighmany`, with it it adds nothing. Two answers to one question are an override, not a composition.
- *The tie at $v_p(b) = 1$.* When p divides the leading coefficient exactly once and not the next coefficient, a denominator with $v_p(b) = 1$ admits a single class of numerators modulo p , which the sieve could use with $p - 1$ further tables per prime. The estimate assumed the prime is in the first stage; it usually is not, because the class it cannot sieve makes it look weak to the ranking. Refined to 0.4 to 1% of a run and not built.
- *Blocking the prime loop for the cache and a two-copy table layout* were proposed for cores with a 32 KB first-level cache or with 256-bit loads executed as two halves; on this machine's performance cores they lose, and for its efficiency cores an analysis of the loop predicted gains of 9% and of up to 46% of the first stage. Their sign depends on the cache size, so they belong behind a run-time gate and a measurement on such a machine, and were not built.

A.7 How much of it is this machine

Every gain this version added is general: it comes from the mathematics (fewer numerators or denominators to look at, fewer rows to build) or from the instruction stream (fewer instructions, fewer mispredicted branches, a shorter dependency chain), and every out-of-order core rewards these in the same direction. An algorithmic gain has the same size everywhere; an instruction-stream gain scales with the core's throughput and its misprediction penalty, which are within a factor of two of each other across current cores. The machine lives in two places. The first is what this version inherited: the register width, the sixteen registers of a chunk (AVX-512 and ARM64

have thirty-two) and the replication of the tables, the last two decided on a core with a 48 KB first-level cache where the mainstream x86 cores of the last decade have 32. The second is the constants: the `COST` values are ratios between different kinds of work — scalar integer code for a table row, a vector load for an AND, branchy code for a survivor — and their unit depends on the register width, so that a 64-bit build has them 1.6 times too small and a 128-bit build 1.2 times; that is inside the flat basin, and `make tune` covers the five that matter. Only the `TEST` macros are intrinsics; the arithmetic on bit arrays is the compiler's vector extension, so a port to another vector instruction set is one more branch of some forty lines in `rp-private.h`.

A.8 Traps for whoever changes the program

- Bit `j` of `den_bits` is the class $b \equiv j \bmod 64$, not $b - 1$, although the set-up shifts by $(b_{\text{low}} - 1) \bmod 64$: the loop shifts before it tests. Read the other way, it inverts the parity for a third of the curves and makes the run-length estimate 2.4 times too small, which looks like a modelling error.
- Changing the number of moduli in the middle of a run is safe for correctness — sieving only removes numerators that cannot be points — but not for the bookkeeping: the list of moduli must hold every informative one, the list of reduced denominators must be sized for all of them, and the buffer of sieving tables must cover every prime looked at, since a prime the third stage held may be promoted and suddenly build a table. Missing the last is a segmentation fault.
- Anything added to the sieve entry structure is read in the innermost loop of the first stage; eight more bytes cost about 1.5% of a long run.
- The sixteen-register ladder of the first stage stays hand-written: as an array of sixteen registers `gcc` spills it, 15% slower. The tail legs need `#pragma GCC unroll` to keep their smaller arrays in registers, and a variable that is in fact a constant must be written as one — an offset written as `t & ~15` instead of 0 keeps eight index registers live and spills five per prime.
- `prime[]` holds the odd primes only, so `prime[0]` is 3.
- With `PRIME_SIZE 9` the mask that records which primes a modulus involves runs out of bits at the 64th prime, so beyond it the exclusion of moduli sharing a prime silently stops; the points are unaffected, the sieve merely does redundant work.
- The output of `-x`, the survivors without the exact check, depends on which moduli were chosen and so on the tuning; only the points are invariant, and a test of the sieve must pin the primes.

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